

STELLAR SPECTRAL CLASSIFICATION

I. APPLICATION OF COMPONENT ANALYSIS

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Summary

The statistical technique of component analysis is investigated as a possible technique for the formulation of classification parameters from spectrophotometric measurements on stellar spectra. A specific case of the late-type giants is discussed as an example, and classification parameters, produced by this method, are given for 84 G and K giants (class III). The technique is attractive in not only giving the formulation of the parameters, but also indicating how many parameters are necessary to describe the sample of stars. Some extensions of the standard component analysis technique are discussed, particularly with regard to the analysis of non-linear correlations and the treatment of incomplete data. Finally, the relation between this method and the more direct method of determination of transformations between a set of measurements and a standard classification system is discussed. Extension of the component analysis method to other astronomical fields is strongly indicated.

1. *Introduction.*—The application of narrow-band photoelectric photometry to the problems of spectral classification has shown considerable advances in the last few years, and such techniques should yield very valuable quantitative information. The present paper is concerned with methods of analysing such measurements (or any analogous set of measurements on a large number of objects), when the primary aim is to set up a system of classification.

Analysis of such measurements is normally with reference to a well known and well established classification system such as the MK system, or to the well-established and precise photoelectric colour systems in use today. When discrepancies arise in such comparisons, it is always difficult to decide its source (assuming that errors of observation have been adequately taken care of), particularly whether an individual star is to be regarded as “peculiar” in the measurements involved, or whether the calibrating classification is in error. The existence of unexplained discrepancies or perhaps a generally larger scatter than anticipated sometimes tempts one to consider the introduction of further parameters of classification; currently two stellar classification parameters are well established—spectral type and luminosity class—and a third, “population”, is in the process of being systematized, although this process is not easy. This situation prompts one to ask whether there is any systematic way of analysing an arbitrarily large set of measurements on a large number of objects so as to ensure that just a necessary but sufficient number of classification parameters is produced.

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In this paper we shall be mainly concerned with a possible technique for such an analysis, and results of purely astronomical interest will be reserved for a later paper, except that, for the purposes of illustration, a specific set of observational data is analysed here and applied to the problem of classifying the late-type giants.

2. *Component analysis*.—There are two possible basic approaches to the problem of how to set up a classification scheme for any set of objects. One may first decide that a limited set of fundamental quantities is of significance for determining the measurable properties of each object. One may then attempt to estimate values (perhaps parametrically) for each of these fundamental quantities from the measurements it is possible to make. The validity of such a procedure depends on the correctness of our ideas as to the basic physical constitution of the object and the associated fundamental quantities. This might be described as a theoretical approach. The opposite “empirical” approach is to take all the measurable aspects of the object and to ask what form of description can be devised which will, purely formally, account for the measured properties.

Symbolically, the first case is represented by saying that there are n distinct fundamental parameters ξ_s , so that for each of m measurable quantities x_t per object there exist m relations:

$$x_t = x_t(\xi_s) \quad s = 1 \dots n, \quad t = 1 \dots m, \quad (1)$$

whose functional form is known or can be determined. Then the problem of classification is that of inverting the transformation to give n relations:

$$\xi_s = \xi_s(x_t) \quad s = 1 \dots n, \quad t = 1 \dots m, \quad (2)$$

from which the classification parameters ξ can be derived, given the measurements. If there is to be any value in classification at all, then we would require that the number of fundamental parameters ξ should be less than the number of *measurable* quantities per object, i.e., $n < m$. (In a practical problem, of course, we should try to restrict the number of quantities *actually measured* per object for the purposes of classification to as near n as possible.)

In the second approach, we do not know the functional form of the equations (1), but are interested in discovering whether there is any set of $n (< m)$ parameters ξ_s , which can be derived from the measurements, such that an inversion of (2) to give (1) is possible, showing that the parameters give a complete description. The historical development and current status of stellar classification is something of a mixture of these two approaches since theoretical and empirical studies tend to advance together, but it is proposed in this paper to take the extreme empirical standpoint and to look for classification parameters solely within the system of narrowband measurements we are considering. Afterwards, however, we shall use some of the known classification properties of stars as a comparison. The technique of component analysis, to be described here in reference to an astronomical application, is one that has been widely known and used for some time by statisticians in the field of biometry and human psychology. A much fuller discussion is given by Kendall (1957) in an interesting book, the whole of which is devoted to this and related techniques of statistical analysis.

We have to find some systematic method of looking for functions of the x_t like equation (2) which can conveniently be inverted to give (1). In order to do this, the assumptions we shall make are: (a) that the ξ_s are related to the x_t by a

linear transformation. If x_{ij} represents the measurement of the i th quantity on the j th star, then we shall assume that the k th parameter for the j th star, ξ_{kj} , is given by

$$\xi_{kj} = \sum_{i=1}^m a_{ki} x_{ij} \quad k=1 \dots m, \quad j=1 \dots p, \quad (3)$$

where p is the number of stars, and the a_{ki} are constant coefficients. It will prove convenient to adopt a matrix notation for this work, so we shall write Ξ , A , and X for ξ_{kj} , a_{ki} and x_{ij} and then equation (3) can be written

$$\Xi = AX \quad (4)$$

and when this is inverted, as in going from equation (2) to equation (1), we get

$$X = A^{-1} \Xi. \quad (5)$$

(b) If we are considering a set of observations x_{ij} which are in a standard form, i.e., which have zero mean and unit variance, then we need only consider, as far as later work is concerned, transformations which are orthogonal, so that $A^{-1} = A'$. In this case $n=m$, and these transformations are equivalent to a rotation of the co-ordinate axes. This fact is so useful that we shall assume that $n=m$ in what follows. If the effective dimension of a classification scheme is less than m (it cannot be greater if we are to make any sense at all of the measurements), then this will become apparent in an interesting way in what follows.

In addition to our assumption that the ξ 's and x 's are related by a linear transformation, we impose the condition that the derived ξ 's are to be statistically independent, i.e., uncorrelated and we shall also take it that the x 's are in a standardized form and that A is orthogonal. Then these conditions are equivalent to asking that the directions of the ξ axes in the n dimensional space be those which minimize the sum of squares of the perpendicular distances of the p points. Kendall, in the reference cited, proceeds in this way by imagining the m observations on each of, say, p objects as representing a distribution of p points in an m -dimensional space, and then asking for the direction cosines of the lines in the space which have a minimum sum of squares of perpendicular distances from each point to the line. It follows from this that the corresponding ξ 's are uncorrelated, and that the transformation matrix is orthogonal.

In this form the problem can be solved in the following way. Suppose we have observations of m quantities on each of p stars. Then we have an "observation matrix" $X = x_{ij}$, $i=1 \dots m$, $j=1 \dots p$, and since the x 's are assumed to be standard normal variates, it follows from the definition of *covariance* that the $m \times m$ matrix of covariances $C = c_{ij}$ is $1/p \sum_j x_{ij} x_{kj}$ and so is the product of X and its transpose X' since $x_{kj} = x_{jk}'$. Then the condition of zero covariance between the ξ 's to give statistical independence implies that we want an orthogonal matrix A such that $\Xi = AX$, and that $K = 1/p \Xi \Xi'$ is *diagonal*. We have

$$\begin{aligned} K &= \frac{1}{p} (\Xi \Xi'), \\ &= \frac{1}{p} AX (AX)', \\ &= \frac{1}{p} AXX' A', \\ &= ACA', \end{aligned} \quad (6)$$

or $KA = AC$ since $A'A = 1$ for an orthogonal matrix.

This leads to the characteristic equations of the matrix C for, if K is to be diagonal, we concentrate attention upon its leading element, which we call λ , and we write (6) in terms of this component only.

$$\lambda a_{1j} = \sum_{k=1}^m a_{1k} c_{kj}. \quad (7)$$

Now if we write α for the vector a_{1k} this may be written

$$\alpha(C - \lambda I) = 0 \quad (8)$$

where I is the unit matrix δ_{kj} . Equation (8) can be solved for α if λ is known, but the condition that it has a solution with the components of α *not all zero* is

$$|C - \lambda I| = 0 \quad (9)$$

which is the well known characteristic equation of a matrix C .

Equation (9) gives an m th order equation for λ which can be solved in terms of the covariances C_{ij} to give m values for λ , the *eigenvalues* of the matrix C . Substitution of one of these into (8) and solution for α gives one *eigenvector* and the set of m eigenvectors forms the required transformation matrix A . Essentially the problem is now solved for a given practical case although the numerical work involved in solving (9) in general for large m may become difficult. Note that, because of the standardization, C is identical with the correlation matrix R of the data.

It is clear from the derivation just given that the eigenvalues form the principal diagonal of the transformed covariance matrix and so are the simple variances of the parameter ξ_s . Hence the parameter corresponding to λ_1 , the largest eigenvalue (i.e., the quantity $\xi_1 = \sum_i a_{1i} x_i$), is the one with the largest principal axis, that is, it represents the largest component of variation. (It is clear physically, and is also proved by Kendall, that this largest principal axis is also the one with the smallest sum of squares of perpendicular distances of individual points. He also proves that the axis corresponding to the next largest value of λ has the next smallest sum of squares, and so on.) Hence *the size of each eigenvalue is a measure of the relative contribution of the associated parameter, ξ , to the total variation within the system of measurements*. A zero eigenvalue indicates that there is one component with no contribution at all, that is, it shows that the effective dimension of the system is less than m . Thus by examining the sizes of the eigenvalues, and comparing them with the expected observational errors, it is possible to determine the dimension of a necessary and sufficient scheme for a description of the system. The test for significantly large eigenvalues has been made in this paper by assuming that, if δx_i represents the average r.m.s. error on the i th observation per star, then the quantity

$$\frac{1}{\lambda_k} (\delta \xi_k)^2 = \frac{1}{\lambda_k} \sum (a_{ki})^2 (\delta x_i)^2 \quad (10)$$

follows the F distribution. This cannot be strictly true since the a_{ki} themselves are derived from the observations and so are subject to errors, but for fairly large amounts of data it should not be too far wrong.

To summarize, we can find parameters which are linearly related to the observed quantities, which are statistically independent, and whose relative (statistical) importance can easily be determined. As a consequence we can reject those

parameters which contribute nothing to the description of the system, and so we can discover its essential dimensionality. The main point of this paper is to suggest that these parameters ξ be used, just as they stand, as classification parameters for the system.

It is important to note that if we had in mind, say, a two-parameter classification scheme, then while *any* two measured quantities not functionally related to one another *could* be used and the resulting classification scheme would be perfectly *internally* consistent, this would not normally be a very useful scheme in practice. If we require to *prove* that our two parameters have some significance, then we have to measure more than two quantities per star. In the notation used earlier, we require $m > n$. The attraction of the principal component method is that it not only gives a "formula" for producing the parameters but it also indicates whether they are really significant, thus giving the required proof. The ideal situation with $m \gg n$ is perhaps difficult to realize in practice since we would not only have to measure many quantities on a new star in order to set up the classification scheme, but also would need to measure as many quantities on a new star in order to classify it on this scheme, each time we wanted to apply the method.

The practical computation necessary to carry out the analysis described in the preceding paragraphs has been performed on the Cambridge University EDSAC-2 computer and on the CDC 1604 computer of The University of Texas. A programme has been developed which will read into the computer a large multi-variate set of data, such as the set of narrow-band observations we are discussing here, and will form the covariance matrix and the means and variances of each set. The computer will then form the eigenvalues and eigenvectors of the covariance matrix. If the data is read in again, then the matrix of eigenvectors is used to transform the data so as to produce the parameters in accordance with (3), and the values of these parameters are output.

Provision is made in the programme for modifying the data while it is read into the machine, so that one can pre-transform it in an arbitrary way before the principal component analysis is carried out. In this way, one can take account of possible known non-linear relations between the x 's and the ξ 's. However, there is an infinite number of possible non-linear relationships which could be used. The development of the programme in such a way as to take account of non-linear correlation is discussed below in Section 4.

We discuss here a set (called set K) of measurements consisting of the Ca II K line (x_5), the 4200 CN band (x_1), the Mg b lines (x_2), H α (x_4), and the red Ca I triplet, 6102–6162 (x_3), in 84 giant stars. This choice of data is a compromise between the requirement of having m/n as large as possible, and that of having as many stars to deal with as possible so that both the statistics become more reliable, and the number of stars for which classification parameters are finally produced is large enough to be useful. With the exception of the K line results which are by Gyldenkerne (1955) all these results are those obtained with the Cambridge narrow-band equipment on the 36-inch telescope. Some of these results have already been published (Griffin & Redman 1959, Deeming 1960), others will appear shortly. The eigenvalues of the resulting 5×5 correlation matrix were 3.281, 0.762, 0.162, 0.485, 0.311. On assuming average errors of 1 per cent on the input data, we find that only the eigenvalues 3.281, 0.763 and 0.162 are significantly greater than the errors (the last only just significant). The

reason why these do not correspond exactly to the largest apparent eigenvalues is that, because of different total ranges of variation of the fundamental measurements, some have relatively larger errors in the normalized form than others. In particular, the $H\alpha$ measurements have a rather large r.m.s. error relative to the total range of variation of the strength of $H\alpha$ in these stars. The parameters with apparent eigenvalues 0.485 and 0.311 happen to have rather large coefficients of $H\alpha$ in them, so they are more sensitive to small errors in $H\alpha$ than the parameter with eigenvalue 0.162, which consequently is the more significant. Note that the sum of the eigenvalues which are significant is 4.205; thus the three most significant parameters contain a little over 80 per cent of the total variation, the rest of which must be presumed to be due to observational errors and residual "cosmic scatter".

It is now clearly of great interest to see whether the parameters, K_i (called K to identify this particular set of data), corresponding to those significant eigenvalues, and derived in a completely empirical way, can be related to known spectral types or to other information about the same stars. Since K_1 has an eigenvalue so much larger than those of K_2 and K_3 , and since we are considering only giant stars with a restricted range in absolute magnitude, it is reasonable to attempt to identify K_1 with a spectral type parameter. Figure 1 is a typical diagram showing the correlation between MK type, and the parameter K_1 . The actual analysis on which this diagram is based is a non-linear analysis to be described later. Clearly the correlation is sufficient to justify our belief that K_1 is to a large extent a measure of spectral type. If we take the step of making a complete identification of K_1 with spectral type (and such a step is to some extent justified by the statistical significance associated with K_1) then it follows that we could produce spectral types from K_1 with a precision rather greater than the MK system—certainly to better than 0.5 of one tenth of a spectral class.

At present, there seems to be no clear interpretation of the other two apparently significant parameters. They are not correlated with published spectroscopic absolute magnitudes for these stars, nor with space velocities. Of course, we should remember that K_3 , in particular, adds only very little to the total variation, so that attention should really be concentrated upon K_2 . We have also restricted attention to giant stars, so that the majority of absolute magnitude variation which some of the measured features are known to show has been eliminated.

Gyldenkerne (1961) has reported the derivation of some classification parameters from narrow-band measurements by methods based more directly upon the calibration of the measurements against known parameters. The degree of correlation of K_1 with values of S (kindly supplied by Dr Gyldenkerne in advance of publication) is very good, and is indeed better than the correlation between K_1 and MK spectral type, thus giving some additional confirmation of the significance of K_1 . We should not, however, put too much weight onto this good inter-agreement, since the set of Ca II K line measurements is in common to the two systems so some degree of correlation would be inevitable. It does not appear to be directly possible to relate K_2 and K_3 to Gyldenkerne's L and C parameters with any degree of statistical significance.

It should be made clear that the correspondence of parameter K_1 with MK types or with the spectral type parameter S is to some extent fortuitous in the sense that it is largely a result of the sample of stars chosen. That is, by restriction to giants we have artificially created a sample of stars in which by far the largest

contributor to variations in the spectrum is temperature. What we have done is to take a sample of stars with effectively just one important factor responsible for spectral variations. We have then shown by our analysis that there is indeed just one important factor which can describe the system of measurements, and have obtained an estimate of this parameter. The test we have made is then to compare this with what we already know is an adequate parameter to describe this essentially one-dimensional situation. If our sample of stars were more closely of a two-factor character, where the factors were equally important, then we would still get a correct description of the system, but the parameters would not necessarily correspond directly with any other (equally valid) two parameter system, say ρ_1, ρ_2 . However, in this case a *functional relation*

$$\begin{aligned}\rho_1 &= \rho_1(\xi_1, \xi_2) \\ \rho_2 &= \rho_2(\xi_1, \xi_2)\end{aligned}\tag{11}$$

would hold between any two adequate classification systems, although it might be that one of these systems would have a more direct physical interpretation and would be preferred on those grounds.

3. *Limitations.*—There are several limitations in the technique as so far described. Some of these are of an essentially practical nature, and in this section we shall discuss some possible methods of dealing with them.

3.1. *Missing observations.*—A very common situation in dealing with large amounts of numerical data, is that certain observations are missing, for one reason or another. On the philosophy of extracting as much information as possible from the available measurements, a technique has been devised for including in the analysis such objects which have some missing observations. In effect, here, if an observation is missing, the programme will make the best estimate as to its value by taking the intercorrelations of the various sets of measurements as found out for all the other stars, and using these as a guide. If measurement x_k is missing for a particular object, its value is taken as:

$$x_k = \bar{x}_k + \sigma_k \frac{1}{n} \sum_m^{n'} r_{km} \frac{x_m - \bar{x}_m}{\sigma_m}\tag{12}$$

where the summation is taken over those values of m corresponding to the n' measurements which are present. The other symbols have the usual definitions. The use of such an estimate is, of course, only necessary at the stage where the final output parameters are produced, and it does not affect the analysis up to that point, nor does it affect the output parameters for any other star. Therefore, if there is any reason to suppose that for a particular star this will not be a good approximation, the results for that star may be rejected without biasing the remainder of the results.

3.2. *Non-linearity.*—A second important problem is that of non-linear correlations, since these occur very frequently, particularly in connection with measurements on stellar spectra. It is convenient to divide this problem into two fields, one of which is much easier to deal with than the other. First, we suppose that the correlations are non-linear, but that for each regression line $x_i = x_i(x_j)$,

x_i is a monotonically increasing function of x_j and *vice versa*. In other words, whatever the values of i and j , i.e., whichever two sets of measurements we choose, the specification of an x_j or an x_i will uniquely determine a point on the regression line, even though the line is not straight. We call the analyses under this assumption "Type I". In this case, an approach which has been adopted here is to pick one of the sets of measurements as a kind of standard set, and to make a preliminary non-linear fit of the other sets of measurements to it by a least squares solution for the constants c_n in the set of equations:

$$x_s = \sum_{n=0} c_n x_k^n, \quad (13)$$

for all $k \neq s$. Here x_s represents the measurement chosen as standard. Then for all sets other than set s , the transformation defined by the right-hand side of (12) is applied to each measurement on each star. It will be clear that the inter-correlation between these transformed measurements will be linear, and that furthermore, none of the essential variations in the original data is lost. Hence the remainder of the technique may now be followed through exactly as before. The present programme will allow terms up to and including x^9 to be used in the transformation, although in practice inclusion of terms up to and including x^4 seems to be perfectly adequate; and in this case, the results are less sensitive to rounding-off errors in the computer. A check on the operation of this process may be made by using terms only up to x^1 , since then we are back to linear transformations, and we should reproduce the results of the original analysis as described in the previous paragraphs, which should, of course, be independent of the choice of the standard. This is indeed found to be the case.

TABLE I

Eigenvalues determined from linear analysis, and all possible non-linear (Type I) analyses, using terms up to x^4

Eigenvalues	λ_1	λ_2	λ_3	λ_4	λ_5
Linear	3.281	0.7619	0.4849	0.3107	0.1618
$s=1$	3.201	0.6810	0.4842	0.3383	0.2950
2	3.278	0.7846	0.4712	0.2937	0.1726
3	3.333	0.7355	0.4861	0.2620	0.1837
4	3.381	0.7369	0.4803	0.2325	0.1692
5	3.371	0.7520	0.4792	0.2459	0.1518

An obvious disadvantage of this method is the assignment of a privileged position to one of the sets of measurements. Here is a case where the judgment of the investigator has to be called upon as to the most suitable choice. In the absence of any available guide as to the choice to be made, a possible approach would be to repeat the whole analysis for all possible choices of standard, and to pick the one which maximizes the largest eigenvalue.

Table I shows the effects of a non-linear analysis of the data according to the scheme described above, with the transformation (13) taken up to terms in x^4 , and for all possible choices of standard x_s . The case $s=4$ gives the largest first eigenvalue, and so we might choose this as standard. One may argue that what one should do is to choose the standard which concentrates as much of the variation into those parameters which are significant. This means, however,

TABLE II

Star	MK Type	$K_1(\text{Linear})$	$K_1(s=4)$	$K_1(s=5)$	$K_2(s=5)$
HD					
34559	G8	-2.222	2.207	2.174	-0.887
37160	G8	-3.372	4.304	3.704	2.955
37984	K0	-0.985	1.013	1.034	0.799
38656	G8	-2.248	2.489	2.401	0.629
38751	G8	-1.099	1.053	1.085	0.099
39003	K0	0.744	-0.746	-0.720	-0.854
40035	K0	-1.740	1.753	1.570	0.706
41597	G8	-1.699	1.659	1.694	0.127
43039	G8	-1.045	1.048	1.151	1.079
47174	K3	0.834	-0.878	-0.797	-0.818
48432	K0	-1.498	1.645	1.674	1.008
48433	K1	0.746	-0.863	-0.863	0.218
48781	K1	-0.194	0.280	0.276	-1.213
49293	K0	0.305	-0.166	-0.156	-1.544
49520	K3	1.145	-1.290	1.305	-0.172
52960	K3	1.748	-1.480	-1.843	0.392
54719	K2	2.712	-2.719	-2.819	-0.907
55280	K2	0.186	-0.362	-0.276	0.727
57264	G8	-0.861	0.876	0.954	0.801
57669	K0	0.470	-0.434	-0.577	-1.478
57727	G8	-3.122	3.578	3.029	0.809
58207	K0	-0.528	0.464	0.553	-0.181
59294	K2	1.961	-1.897	-2.175	-0.761
62509	K0	-0.795	0.681	0.740	-0.623
66216	K2	1.422	-1.624	-1.629	0.286
74442	K0	1.131	-1.291	-1.261	-0.034
75506	K0	-2.436	2.457	2.623	0.323
77912	G8	-1.857	1.629	1.575	-1.154
78515	K0	-0.581	0.546	0.617	-1.037
81146	K2	2.523	-2.704	-2.734	-0.314
83805	G8	-1.340	1.291	1.358	-0.729
85503	K2	3.924	-3.735	-3.791	-0.421
92424	K2	2.514	-2.499	-2.369	-0.474
92523	K3	1.788	-1.842	-1.821	1.793
94247	K3	2.532	-2.644	-2.655	1.122
94264	K0	-0.083	-0.037	0.043	0.644
94600	K1	0.091	0.014	0.096	0.153
95689	K0	-0.919	0.798	0.879	-0.814
96833	K1	0.147	-0.240	-0.168	-0.129
97907	K3	0.218	-0.371	-0.224	0.538
98262	K3	1.639	-1.751	-1.596	1.123
100696	K0	-0.811	0.684	0.792	-0.066
101484	K0	-1.027	0.999	1.107	-0.399
102224	K0	0.393	-0.503	-0.429	0.613
102328	K3	3.271	-3.219	-3.181	0.494
104979	G8	-2.812	3.129	2.919	0.229
106714	K0	-2.081	2.165	2.293	0.391
106760	K1	0.432	-0.568	-0.474	0.408
107328	K0	-0.098	-0.011	0.014	0.143
107383	G8	-1.964	2.038	2.153	0.166

TABLE II (*continued*)

Star HD	MK Type	$K_1(\text{Linear})$	$K_1(s=4)$	$K_1(s=5)$	$K_2(s=5)$
107950	G7	-3.036	2.036	2.877	0.197
108225	G8	-2.162	2.114	2.036	-0.305
109317	K0	-1.072	1.042	1.169	-0.618
113092	G8	0.372	-0.362	-0.240	0.919
113226	G9	-1.767	1.484	1.588	-1.172
115004	K0	0.001	0.039	0.079	-1.474
120539	K4	2.554	-2.497	-2.683	1.579
121710	K3	2.776	-2.476	-2.915	0.951
124679	K0	-0.771	0.709	0.808	-0.447
124897	K2p	1.216	-1.457	-1.346	1.160
125560	K3	3.556	-3.459	-3.485	-0.040
127665	K3	2.782	-2.734	-2.787	0.294
129312	G8	-0.967	0.843	0.922	-1.576
129972	K0	-1.028	0.750	0.894	-1.230
131111	K0	-0.651	0.587	0.669	0.573
133124	K4	2.922	-2.428	-2.714	1.734
133165	K0	0.471	-0.569	-0.482	-0.401
133208	G8	-1.484	-1.264	1.367	-1.210
133582	K2	1.573	-1.706	-1.663	-0.156
134190	G8	-2.053	1.748	1.873	-0.433
135722	G8	-1.977	2.091	2.165	-0.323
137759	K2	2.845	-1.957	-1.793	-0.272
139641	G8	-3.069	4.003	3.664	1.477
140573	K2	2.679	-2.749	-2.711	-0.959
142091	K0	0.710	-0.940	-0.863	0.517
143107	K3	1.628	-1.762	-1.727	-0.048
143666	K0	-1.357	1.326	1.448	-0.379
145328	K0	0.178	-0.272	-0.314	0.228
147677	K0	-1.084	1.022	1.094	-0.864
150997	G8	-2.659	2.470	2.536	-0.084
153210	K2	2.240	-2.316	-2.439	-0.630
168775	K2	1.697	-1.682	-1.722	-1.059
169414	K2	1.194	-1.338	-1.397	0.178
181276	K0	-1.566	1.444	1.541	-1.229

TABLE III

Mean K_1 parameter for MK types, to be used for interpreting K_1 as a spectral type

MK	$K_1(s=5)$	MK	$K_1(s=5)$
G6	4.25	K1	-0.40
0.5	3.80	0.5	-0.85
G7	3.35	K2	-1.30
0.5	2.85	0.5	-1.80
G8	2.40	K3	-2.20
0.5	1.90	0.5	-2.70
G9	1.45	K4	-3.15
0.5	1.00	0.5	-3.65
K0	0.55	K5	-4.10
0.5	0.05		

a prior knowledge as to how many parameters are to be used. If, guided by the previous results, we suppose that only two parameters will be significant, then we should choose the value of s which maximizes the sum $\lambda_1 + \lambda_2$. This turns out to be $s = 5$. If we go to three parameters, we also find that $s = 5$ gives the maximum value of $\lambda_1 + \lambda_2 + \lambda_3$.

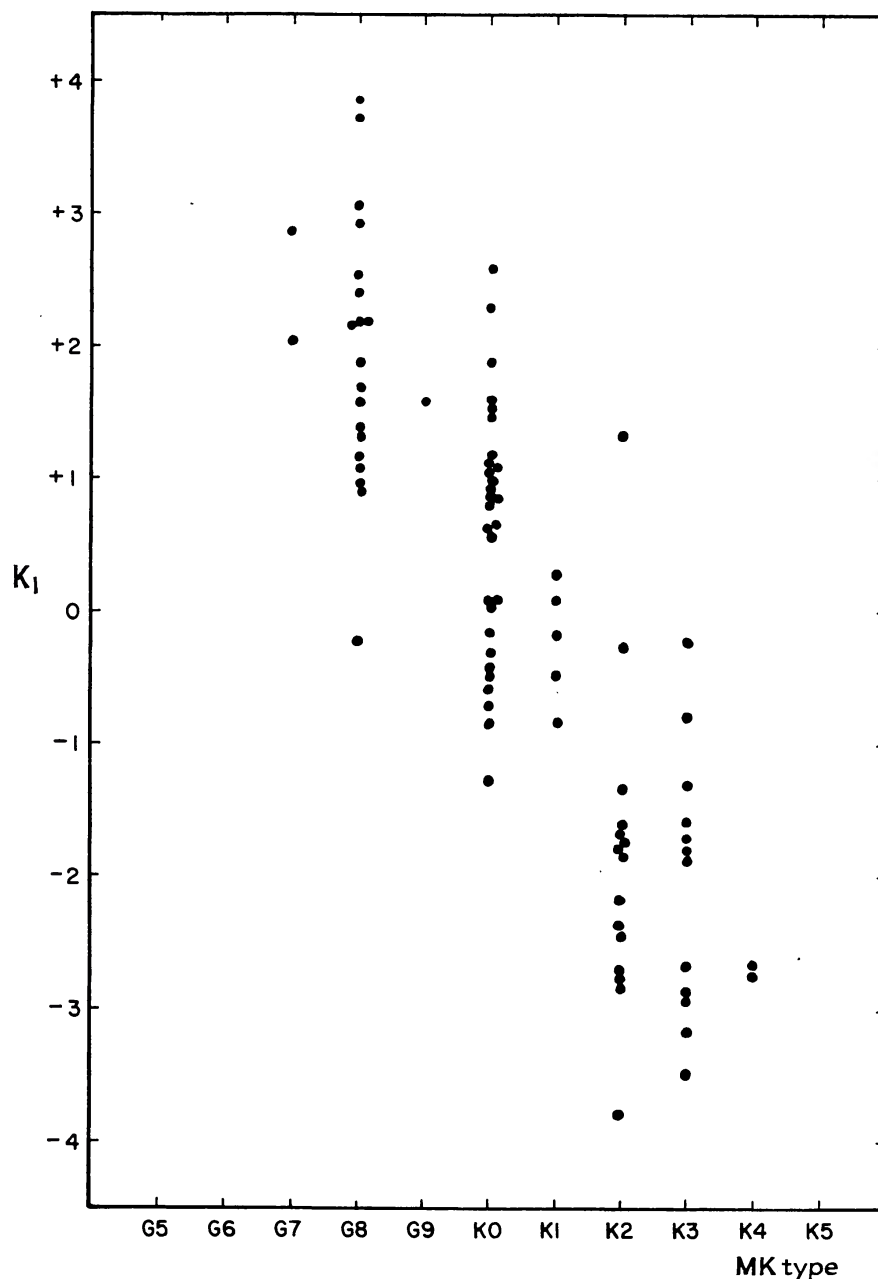


FIG. 1.—Comparison of K_1 ($s=5$) with MK types.

On the basis of these results, it seems that the parameters derived from a non-linear analysis (Type I) with $s=5$ probably represent the “best” that we can do here. (Changes in the degree of the fitting curve cause no significant variation in the results.) Table II shows the parameters, K_1 for the linear analysis and for $s=4$ and $s=5$, as well as the parameter K_2 for $s=5$. Table III shows the

corresponding mean values of K_1 ($s=5$) for various MK types—these may be used to interpret the parameters K_1 ($s=5$) in Table II in terms of corresponding MK types, for reclassification of these stars. Figure 1 shows the correlation between parameter K_1 for $s=5$, and MK types. The correlation coefficient in this diagram is 0.86. If we allow for errors of one tenth of a class in the MK types, this will be corrected to 0.91. We now regard the parameters K_1 and K_2 as our best estimates for a two-parameter classification system for this sample of stars.

Table 4 and 5 show the full correlation matrix, and its eigenvalues and eigenvectors for this case $s=5$. This will be used for illustrative purposes later.

TABLE IV

Correlation matrix

1.000	0.285	0.546	0.463	0.497
0.285	1.000	0.700	0.603	0.812
0.546	0.700	1.000	0.558	0.761
0.463	0.603	0.558	1.000	0.615
0.497	0.812	0.761	0.615	1.000

TABLE V

Eigenvalues and eigenvectors

$k =$		1	2	3	4	5
λ_k		3.371	0.752	0.479	0.246	0.152
a_{ki}	$i=1$	0.354	0.850	-0.110	-0.277	-0.251
	2	0.463	-0.496	-0.085	-0.288	-0.670
	3	0.479	0.013	-0.403	0.779	0.037
	4	0.429	0.022	0.879	0.187	0.088
	5	0.497	-0.175	-0.212	-0.446	0.692

3.3. *Multivalued correlations.*—The problem of what happens when there are multivalued correlations between the measurements ("Type II" problem) is a good deal more difficult, but could perhaps be handled in a similar kind of way. Again it would be necessary to pick one measurement as a standard, and here a considerable degree of good judgment would be called for. Let us suppose that we can find a standard set s such that the relations between the other sets of measurements, x_k , and the set x_s , are multivalued only in the following restricted way: a given value of x_s will uniquely determine a corresponding x_k on each of the mean relations, although a given value of x_k will not normally determine a unique value of x_s . Perhaps the best known example to make this clear is that of the well known two colour diagram for the Hyades main sequence stars. Here, because of the influence of the Balmer discontinuity, the value of $U-B$ goes through a minimum at around $B-V=0.45$, rising to a maximum at around $B-V=0.2$. Within the range $0.0 < B-V < 0.55$ (approximately), a given value of $B-V$ will uniquely determine a corresponding value of $U-B$, although a given value of $U-B$ will give three equally good alternatives for $B-V$. Thus $B-V$ here would serve for x_s , whereas $U-B$ would not. In accordance with the technique used before, we might fit each measurement to the set x_s by a curve

$$x_k = x_k(x_s). \quad (14)$$

The form of this assumes the requirements we placed on x_s , which can be alternatively stated that x_k must be expressible as an explicit function of x_s . Again in analogy to what we did before, we need to replace each x_k by an x_s obtained by solving (14) and to choose that x_s which is closest to the observed x_s for that star, in order to get onto the correct part of the curve. Then each observation x_k is transformed into something which contains all the information and intrinsic variation as the original x_k , but which now is correlated linearly with the other transformed measurements. From here on the analysis proceeds as before. The one great difficulty associated with this, however, stems from the intrinsic scatter of the measurements, which in any case is what we are trying to analyse, for in the region of the maxima (or minima) of the mean curve represented by the above formula, there will be points lying completely above (or completely below) the mean curve and for these it is not possible to choose an x_s which will give the correct x_k . In extreme cases where the degree of correlation involved is relatively small, this will be a serious problem, and it will be necessary to find a way of dealing with these points in a way which does not bias the results. Work is continuing on the best method of dealing with such multivalued correlations.

3.4. *Estimation of parameters.*—If we have decided that only a limited number of parameters, say the first q , out of the m are significant (we can arrange them in any order), then we can ask whether a measurement of only q of the initial input quantities, instead of m , will give an adequately good estimate of the q significant parameters.

First, we may remark that the degree of correlation of any one of the initial sets of measurements x_i with any one of the parameters ξ_k is proportional to the coefficient a_{ki} of x_i in the formula for ξ_k . (We are now talking about the linear analysis for simplicity. For the non-linear analysis, it is only necessary to replace x_i by an appropriately defined $f(x_i)$.) This is intuitively fairly obvious, and in fact it is relatively easy to show that, if we call the correlation coefficient between x_i and ξ_k , $\rho_{k,i}$ (where now the order of the subscripts is important), then we have

$$\rho_{k,i} = \lambda_k^{1/2} a_{ki}. \quad (15)$$

Hence, out of the definition of any parameter ξ_k , we can pick those x_i which are best correlated with it. Essentially the same results may be expressed in another way. Each parameter is defined in terms of the observations by equation (4), and in turn each observation is expressible in terms of the parameters by equation (5). Since A is an orthogonal matrix, we can write equation (5) as

$$X = A' \Xi \quad (16)$$

or

$$\begin{aligned} x_i &= \sum_k a'_{ik} \xi_k \\ &= \sum_k a_{ki} \xi_k \end{aligned} \quad (17)$$

since $a'_{ik} = a_{ki}$.

The x 's and the ξ 's have zero mean, and the ξ 's are uncorrelated, so the variance of x (which is numerically unity) is

$$x_i^2 = \sum_k a_{ki}^2 \lambda_k = 1. \quad (18)$$

Therefore, $a_{ki}^2 \lambda_k$ measures how much of the variance of x_i is accounted for by parameter ξ_k .

Now suppose we find that only the first say q parameters are significant, then (17) becomes

$$x_i = \sum_{k=1}^q a_{ki} \xi_k \quad (19)$$

and it is possible to pick any q of these equations and solve for the significant ξ 's in terms of q of the x 's. It will clearly be most profitable to choose those x 's which are most highly correlated with the significant parameters. Then the significant ξ 's are expressible in terms of the chosen x 's by an inversion of the submatrix a_{ki} (where $i, k = 1 \dots q$) of the total eigenvector matrix A .

TABLE VI

$a_{ki}^2 \lambda_k$					
$k =$	1	2	3	4	5
$i = 1$	0.4224	0.5433	0.0058	0.0189	0.0096
2	0.7227	0.1850	0.0034	0.0204	0.0682
3	0.7733	0.0001	0.0778	0.1493	0.0002
4	0.6203	0.0004	0.3701	0.0086	0.0012
5	0.8326	0.0230	0.0215	0.0489	0.0728

(For explanation, see text.)

A simple tabulation of the quantities $a_{ki}^2 \lambda_k$ is given in Table VI for the data of Table V. This shows directly that x_1 and x_2 both need ξ_1 and ξ_2 to account for their variation, x_3 needs both ξ_1 and ξ_4 , x_4 needs ξ_1 and ξ_3 while x_5 needs only ξ_1 . In view of what we said before, the contributions of ξ_4 and ξ_3 to x_3 and x_4 , respectively, can most likely be interpreted as the effects of observational errors. Thus x_5 would be a good quantity to measure if we only need to know ξ_1 . If we need both ξ_1 and ξ_2 , then probably x_1 and x_5 would be the best, although x_1 and x_2 would do almost as well. In many ways, Table VI gives the most instructive presentation of the way the basic data variation is described by the parameters ξ_k , and shows directly which parameters may be rejected as non-significant, and the effect that such rejection will have on the adequacy of the resulting system as a description of the original data.

4. *Relationship to standard least squares technique.*—There is an obvious similarity between the methods of analysis discussed here and the method of least squares as applied to the fitting of coefficients in a linear equation by the familiar method of forming normal equations and solving them. The most important characteristic of this problem is that there exists a dependent variable, say y , and we are trying to find the best line or hyperplane which relates y to a number of independent variables, x_i , supposed to be known without error. In the component analysis procedure described above there is no such dependent variable. However, it is of interest to discuss the relation between the two approaches, since we may wish to compare the present results with those obtained by, for instance, performing a least squares fit of the MK spectral types to the five sets of measurements used, and also the similarity suggests that there may be other important problems which can be solved by essentially the same approach.

4.1. *Least squares method.*—We shall consider the case of m independent error-free variables x_i , and one dependent variable y , and we require to find the constants b_i , in the equation

$$\sum_i b_i x_i = y \quad (20)$$

(We consider the case where the relations involved are linear, or that the variables represent very small corrections to some first or highest order approximation.) Then each “observational equation” is

$$b_1 x_{1j} + b_2 x_{2j} + \dots + b_m x_{mj} = y_j \quad (21)$$

and according to the standard procedure, we form the normal equations

$$b_1 \sum_j x_{1j} x_{1j} + \dots + b_k \sum_j x_{kj} x_{kj} + \dots + b_m \sum_j x_{mj} x_{mj} = \sum_j x_{ij} y_j \quad (22)$$

assuming that each “observational equation” is of unit weight. If the x ’s and the y ’s are in standard form, then $\sum_j x_{ij} x_{kj}$ is proportional to r_{ik} , the correlation coefficient between x_i and x_k . Hence the normal equations may be written

$$\sum_{k=1}^m b_k r_{ik} = s_i \quad (23)$$

where s_i denotes the correlation coefficient between y and x_i . This may be written in matrix form

$$Rb = s \quad (24)$$

and the solution is

$$b = R^{-1}s \quad (25)$$

so the problem reduces to that of finding the reciprocal of the matrix R . This is the standard approach. There are now two alternative methods of dealing with the problem.

4.2. *First alternative approach.*—First of all, let us return to equation (20) and let us change the notation so that we now write x_{m+1} instead of y , then we are to determine the coefficients in

$$\sum_{i=1}^m b_i x_i = x_{m+1} \quad (26)$$

or

$$\sum_{i=1}^{m+1} b'_i x_i = 0 \quad (27)$$

where the coefficients b'_i are defined by

$$b'_i = b_i \quad \text{for } i = 1 \dots m \quad (28)$$

$$b'_{m+1} = -1.$$

The principle of least squares now requires that the b'_i in (27) be chosen such that the sum of the squares of the quantities represented by the left-hand side of (27), when the observed values of $x_1 \dots x_m$ and $x_{m+1} = y$ are substituted into it, should be a

minimum. It is clear that, with the requirement $b'_{m+1} = -1$, this gives a unique determination of the coefficients b_i . Now suppose we form the $m+1 \times m+1$ correlation matrix

$$r_{ik} = \frac{1}{p} \sum_{j=1}^p x_{ij}x_{kj} \quad i, k = 1 \dots m+1 \quad (29)$$

including the variable y as x_{m+1} , and suppose we find the eigenvalues and eigenvectors of this matrix, λ_i and a_{ik} . The parameter (in the sense used earlier in this paper) which corresponds to the eigenvalue, λ_m , is then defined by

$$\xi_m = \sum_{i=1}^{m+1} a_{mi}x_i \quad (30)$$

and the variance of ξ_m is of course λ_m^* . Now let us express the left-hand side of (25) in terms of the ξ 's rather than the x 's. This can always be done. We write

$$\rho = \sum_m \alpha_m \xi_m \quad (31)$$

where the α_m are to be chosen in such a way as to minimize the variance of ρ , subject to the condition that the coefficient of x_{m+1} in (29) is -1 . Because the ξ_m all have zero mean, it follows that ρ also has zero mean and hence its variance is just its mean square. Since also the ξ_m 's are uncorrelated, and their variances are λ_m , we have for the variance of ρ

$$\overline{\rho^2} = \sum_m \alpha_m^2 \lambda_m. \quad (32)$$

Also

$$\begin{aligned} \rho &= \sum_m \alpha_m \sum_i a_{mi}x_i, \\ &= \sum_i \left(\sum_m \alpha_m a_{mi} \right) x_i, \end{aligned} \quad (33)$$

from which the coefficient of the dependent variable, which for generality we shall label here as s , rather than $m+1$, is

$$\sum_m \alpha_m a_{ms}. \quad (34)$$

Thus our problem is to minimize the quantity

$$F = \sum_m \alpha_m^2 \lambda_m \quad (35)$$

subject to the condition

$$\sum_m \alpha_m a_{ms} = -1, \quad (36)$$

where the a_{ms} and the λ_m are known constants.

To solve this problem, let us pick a particular value of m , say t , then

$$\alpha_t = \frac{-\left(1 + \sum_m (1 - \delta_{tm}) \alpha_m a_{ms}\right)}{a_{ts}}, \quad (37)$$

* m here is just a dummy subscript, not to be confused with the order of the matrix.

and hence

$$F = \frac{\left(1 + \sum_m (1 - \delta_{tm}) \alpha_m a_{ms}\right)^2 \lambda_t}{a_{ts}^2} + \sum_m (1 - \delta_{tm}) \alpha_m^2 \lambda_m$$

$$= \frac{\left\{1 + 2 \sum_m (1 - \delta_{tm}) \alpha_m a_{ms} + \left(\sum_m (1 - \delta_{tm}) \alpha_m a_{ms}\right)^2\right\} \lambda_t}{a_{ts}^2} + \sum_m (1 - \delta_{tm}) \alpha_m^2 \lambda_m. \quad (38)$$

Now minimize F with respect to α_m , i.e., $\partial F / \partial \alpha_m = 0$ for $m \neq t$. This gives

$$\frac{2(1 - \delta_{tm}) \alpha_m \lambda_t}{a_{ts}^2} + \frac{2 \left\{ \sum_m (1 - \delta_{tm}) \alpha_m a_{ms} \right\} (1 - \delta_{tm}) a_{ms} \lambda_t}{a_{ts}^2} + 2(1 - \delta_{tm}) \alpha_m \lambda_m = 0. \quad (39)$$

Since $\delta_{tm} = 0$ for $t \neq m$, this may be written

$$\frac{2a_{ms}}{a_{ts}^2} \left(1 + \sum_m (1 - \delta_{tm}) \alpha_m a_{ms}\right) \lambda_t + 2\alpha_m \lambda_m = 0 \quad (40)$$

and from (37) we get

$$-2 \frac{a_{ms}}{a_{ts}} \alpha_t \lambda_t + 2\alpha_m \lambda_m = 0, \quad (41)$$

or

$$\frac{\lambda_m \alpha_m}{a_{ms}} = \frac{\lambda_t \alpha_t}{a_{ts}} = \dots \quad (42)$$

Hence

$$\alpha_m = \frac{\lambda_t}{\lambda_m} \frac{a_{ms}}{a_{ts}} \alpha_t \quad (43)$$

and since $\sum_m \alpha_m a_{ms} = -1$, we get

$$\frac{\lambda_t \alpha_t}{a_{ts}} \sum_m \frac{a_{ms}^2}{\lambda_m} = -1, \quad (44)$$

or

$$\alpha_t = \frac{a_{ts}}{\lambda_t} \cdot \frac{-1}{\sum_m \frac{a_{ms}^2}{\lambda_m}}, \quad (45)$$

which is the answer to our problem.

What happens if one or more of the λ 's is zero? In this case it follows from (42) that if λ_t is zero, then for $\lambda_m \neq 0$, $\alpha_m = 0$, while for $\lambda_t = 0$, α_t is undetermined by (45). In particular, when only one of the λ 's, say λ_t , is zero, then because $\sum_m \alpha_m a_{ms} = -1$ we have simply

$$\alpha_t = -\frac{1}{a_{ts}} \quad \text{and} \quad \alpha_{m \neq t} = 0. \quad (46)$$

(This special result could have been derived more easily.) However, if there are two eigenvalues which are zero, say λ_u and λ_v , then we have instead

$$\alpha_t a_{ts} + \alpha_u a_{us} = -1, \quad (47)$$

and α_t and α_u have to be chosen to satisfy this equation, so that one of them may be chosen arbitrarily and the other is then expressed in terms of it. Thus the choice $\alpha_u = 0$ in (47) will lead to the same result as (46), but it would also be possible to choose $\alpha_t = 0$, giving

$$\alpha_u = -\frac{1}{a_{us}} \quad \text{and} \quad \alpha_{m \neq u} = 0, \quad (48)$$

which would be just as adequate a solution, and a whole set of intermediate values are also possible. This illustrates an important point, made also by Kendall in the reference cited, that the coefficients in the solution of normal equations are not well determined if the observed variables in each observational equation are highly correlated, implying at least two zero or very small eigenvalues of the matrix r_{ik} in (27).

Unfortunately, these apparently useful properties of zero eigenvalues are of very little practical value since because of observational errors it never happens that a matrix derived from observational data has a strictly zero eigenvalue. It does not appear to be adequate to assume, for instance, that the smallest eigenvalue is effectively zero, even though it may be less than what could be accounted for by observational errors. In other words, this approach would give a biased estimate of the coefficient α_t . As for example, with our 5×5 matrix, let us take the dependent variable as x_5 . Then the assumption that $\lambda_t = 0$ ($t=5$ in this case) leads to $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ and $\alpha_5 = -1.45$ while the exact results derived from equation (42) are

$$\begin{aligned}\alpha_1 &= 0.06 \\ \alpha_2 &= -0.04 \\ \alpha_3 &= 0.43 \\ \alpha_4 &= 0.11 \\ \alpha_5 &= -1.09,\end{aligned}$$

so that the approximation $\lambda_t = 0$ leads to an error of the order of 35 per cent.

This statement is open to argument, since the least squares technique is only supposed to apply to the case where there are no errors on any of the x 's except x_s . If all the x 's are subject to error, then the standard least squares technique does not apply. However, if it turns out that the mean errors on the x 's are all equal (after the x 's have been normalized to zero and unit variance), then one can show that, on the assumption that a true linear functional relation does exist, and that the observed scatter is all due to random errors with the same mean error for all x 's, then the proper procedure is to find the smallest eigenvalue of the *observed* correlation matrix, and to *equate the corresponding parameter to zero*. This equation, $\xi = 0$, then constitutes the desired functional relation between the x 's. When the errors are not all equal, it is not clear what the correct procedure is. It may be that taking $\xi_m = 0$ where λ_m is the minimum eigenvalue is in some sense *better* than the method of least squares, but in what sense is not clear. Obviously it cannot be generally valid, for in the extreme case where all but one of the mean errors is zero, we are back to the standard least squares situation—which we have seen is not equivalent to taking $\xi_m = 0$. If there are only two variables, the problem can be solved, but there is no answer yet to the general problem. Further discussion of this topic here is beyond the scope of the present paper.

From equations (31) and (42), it follows that the coefficients b'_i in equation (25) are given by:

$$b'_i = \sum_m \alpha_m a_{mi} \quad (49)$$

$$= \frac{\sum_m \frac{a_{mi} a_{ms}}{\lambda_m}}{\sum_m a_{ms}^2 \lambda_m} \quad (50)$$

which of course satisfies the condition that $b'_s = -1$. This suggests that it would be useful to define a matrix q_{rs} by

$$q_{rs} = \sum_m \frac{a_{mr} a_{ms}}{\lambda_m} \quad (51)$$

so that if b'_{is} are the coefficients involved in fitting an equation like (20) when x_s is the dependent variable to be fitted, we have

$$b'_{is} = - \frac{q_{is}}{q_{ss}} \quad (52)$$

which is effectively the answer to our problem, equivalent to equation (25) but in a somewhat more useful form. Computation of the matrix Q has been incorporated into the present version of the computer programme, and is given, for the eigenvectors of Table V, in Table VII.

TABLE VII

Q matrix (for explanation, see text)

1.751	0.9420	-0.7810	-0.4874	-0.7405
0.9420	3.705	-0.9489	-0.7184	-2.312
-0.7810	-0.9489	0.2884	-0.0639	-0.9960
-0.4874	-0.7184	-0.0639	1.8610	-0.2712
-0.7405	-2.3120	-0.9960	-0.2712	4.1680

This is largely just for interest, since in the case under discussion here, where we have performed a preliminary non-linear transformation of the data, involving terms in x^4 , the application of equation (52) by manual calculations becomes rather cumbersome. However, the values of Q may be retained in the computer for further use. It happens, as we shall see, that the Q matrix turns up again in a slightly different way in what follows.

Some mathematical properties of Q , easily proved from the definition (51), are:

$$(a) \quad \sum_r q_{rr} = \sum_m \frac{1}{\lambda_m}, \quad (53)$$

$$(b) \quad \sum_i \sum_j q_{ij}^2 = \sum_m \frac{1}{\lambda_m^2}, \quad (54)$$

$$(c) \quad \sum_i q_{ij}^2 = \sum_k \frac{a_{kj}^2}{\lambda_k^2}. \quad (55)$$

4.3. *Second alternative approach.*—Now let us suppose that, instead of including the dependent variable y as x_{m+1} , we merely make an analysis of the $m \times m$ matrix, giving us m eigenvalues λ_i and m parameters ξ_i . Then we attempt to fit the parameters ξ_i by an equation of the form

$$y = \sum_{i=1}^m \beta_i \xi_i, \quad (56)$$

where we again assume y to be adjusted to zero mean and unit variance. Again the β_i are to be chosen so that the variance of

$$\epsilon = y - \sum_{i=1}^n \beta_i \xi_i \quad (57)$$

is minimized. This now gives a relatively simple expression for β_i since

$$\overline{\epsilon^2} = \overline{y^2} - 2 \sum_i \beta_i \overline{y \xi_i} + \sum_i \beta_i^2 \overline{\lambda_i} \quad (58)$$

and for a minimization with respect to β_i we require that

$$\frac{\partial \overline{\epsilon^2}}{\partial \beta_i} = 0 \quad (59)$$

or

$$-2 \overline{y \xi_i} + 2 \beta_i \overline{\lambda_i} = 0 \quad (60)$$

from which

$$\beta_i = \frac{\overline{y \xi_i}}{\overline{\lambda_i}}, \quad (61)$$

which is an analytically simpler form than (23).

Now, letting $\xi_i = \sum_j a_{ij} x_j$, this gives us

$$\beta_i = \frac{1}{\overline{\lambda_i}} \sum_j a_{ij} \overline{x_j y}. \quad (62)$$

If x_j and y both have zero mean and unit variance, and if we denote y by x_s as before, then

$$\overline{x_j y} = r_{js} \quad (63)$$

and

$$\beta_i = \frac{1}{\overline{\lambda_i}} \sum_j a_{ij} r_{sj}. \quad (64)$$

Now

$$\begin{aligned} \sum_i \beta_i \xi_i &= \sum_i \beta_i \sum_j a_{ij} x_j \\ &= \sum_j \left(\sum_i \beta_i a_{ij} \right) x_j, \end{aligned} \quad (65)$$

from which, b_k , as defined in equation (20), is given by

$$b_k = \sum_i \beta_i a_{ik}. \quad (66)$$

If we substitute (64) into (66), we get

$$\begin{aligned} b_k &= \sum_i \frac{1}{\overline{\lambda_i}} \sum_j a_{ij} r_{sj} a_{ik} \\ &= \sum_j r_{sj} \sum_i \frac{a_{ij} a_{ik}}{\overline{\lambda_i}} \\ &= \sum_j r_{sj} q_{jk} \end{aligned} \quad (67)$$

by the definition of the Q matrix. Notice, however, that here we have the Q matrix of an $m \times m$ correlation matrix, whereas before the Q matrix referred to an $m+1 \times m+1$ correlation matrix.

But now, both r_{sj} and q_{jk} are symmetric, hence

$$b_k = \sum_j q_{kj} r_{js}. \quad (68)$$

This may be written in matrix notation as

$$b = Qs, \quad (69)$$

where s denotes the vector r_{js} . Comparison of this with (25) shows that Q is nothing more than the reciprocal of the matrix R ,

$$Q = R^{-1}. \quad (70)$$

The resultant formula for the inverse of R ,

$$(R^{-1})_{ij} = \sum_k \frac{a_{ki} a_{kj}}{\lambda_k} \quad (71)$$

is one which this author has not seen quoted in the standard sources. It is easy to show that it is true for any matrix R which has orthogonal eigenvectors. We shall continue to use the notation Q rather than R^{-1} for the matrix defined by (51).

Comparison of (68) with (52) shows that

$$\sum_{\substack{j=1 \\ \neq s}} r_{sj} q_{ji} = - \frac{q'_{si}}{q'_{ss}} \quad (72)$$

where q refers to the matrix which does not contain the row and column r_{sj} , and q' (not a transpose) refers to the matrix which does contain it. This might be a useful relationship for discussing the general effect of adding a row and column to the correlation matrix, i.e. of including an extra set of measurements in the original data, but (72) of itself does not do this directly.

If we substitute one of the expressions (61), (62) or (64) for β_i back into expression (51) for the variance of the residuals ϵ , we find

$$\overline{\epsilon^2} = \overline{y^2} - \sum_i \beta_i^2 \lambda_i. \quad (73)$$

Thus, the "fitting" of parameter ξ_i reduces the total variance in the residuals by $\beta_i^2 \lambda_i$, which apparently suggests that it is of little use to include parameters for which λ_i is zero or very small, for then the reduction in the variance of ϵ is not great. However, this does depend on the size of β_i , which is large when λ_i is small.

4.4. Application to specific data.—The best way to test the usefulness of these results is to apply it to the case in hand. Thus, we have taken independently estimated MK types for these stars and computed the correlation matrix including this data, and have formed the Q matrix corresponding to this.

Table VIII shows the eigenvalues and eigenvectors of the $m+1 \times m+1$ matrix which includes the row and column r_{sj} , also shown in Table VIII. Table IX contains the Q matrix for the data of Table VIII. From Table IX, it is easy to find the coefficients b_k , using equation (52), with $s=6$. These results are also given in Table IX. b_6 is -1 , as it should be. We can also see that the coefficient

of x_5 is larger than the others, indicating that x_5 (the Ca II K line) is a relatively good estimator of MK type. Of course, as remarked before, practical application of these values is best done within the computer, as a fourth-order non-linear transformation is required to connect the observed Ca II K line measurement with the x_5 appearing in equations (24) or (25).

TABLE VIII

*Eigenvalues and eigenvectors of $m+1 \times m+1$ matrix including
MK spectral types*

λ_k	4.166	0.779	0.482	0.257	0.192	0.125
a_{ki}	0.3009	0.8675	-0.1288	-0.2390	-0.2382	-0.1627
	0.4215	-0.4280	-0.0425	-0.0601	-0.7364	-0.3026
	0.4265	0.06763	-0.3742	0.8071	0.1351	0.0619
	0.3790	0.08846	0.8933	0.1892	0.0537	0.1091
	0.4533	-0.1317	-0.1943	-0.3880	0.0647	0.7646
	0.4482	-0.1858	-0.0763	-0.3185	0.6130	-0.5306

Correlation coefficients with MK types

	$j=1$	2	3	4	5
r_{sj}	0.443	0.789	0.746	0.646	0.861

TABLE IX

Q matrix for data of Table VIII

1.753	0.9287	-0.7930	-0.4978	-0.7798	0.0755
0.9287	3.852	-0.8176	-0.6038	-1.878	-0.8314
-0.7930	-0.8176	3.002	-0.03868	0.0679	-0.7445
-0.4978	-0.6038	0.03868	1.951	0.06753	-0.6498
-0.7798	-1.878	-0.6079	0.06753	5.449	-2.457
0.07547	-0.8314	-0.7445	-0.6498	-2.457	4.714

Coefficients b'_{is} for $s=6, i=1 \dots 6$

	$i=1$	2	3	4	5	6
b'_{is}	-0.0160	0.1764	0.1579	0.1378	0.5212	-1.0000

TABLE X

β_i and $\beta_i^2 \lambda_i$ (for explanation, see text)

	β_i	$\beta_i^2 \lambda_i$
$i=1$	0.4700	0.7448
2	-0.1882	0.0266
3	-0.0650	0.0020
4	-0.1301	0.0042
5	+0.2660	0.0108
		$\sum \beta_i^2 \lambda_i$ 0.7884

In Table X, we have given the values of β_i , computed from equation (64), and $\beta_i^2 \lambda_i$, for $i=1 \dots 5$. The eigenvalues used in (64) refer, of course, to those in Table V, not those in Table VIII. As shown by (73), these results tell us how much of the total variance of y (in this case MK types) is taken care of by fitting parameter ξ_i in (56). These results show, first, that the fitting of parameter ξ_1 takes care of a large part of the variance, and that relatively little extra variance

of y is taken care of by the remaining parameters. We see that $\sum_i \beta_i^2 \lambda_i = 0.7884$, so that the variance of the residual ϵ , after all five ξ_i have been fitted, is, since y is normalized to zero mean and unit variance, $1 - 0.7884 = 0.2116$. Therefore, 21.16 per cent of the total variance of the MK types is unaccountable for by the parameters derived from our observational data, i.e. by all the observational data, since all the derived parameters have been fitted. One might be tempted to identify this residual 21.16 per cent with observational errors in the MK types. The mean error in MK type which would correspond with 21.16 per cent of the total variance of the MK types is about 1.4 tenths of a spectral class—a result not in violent disagreement with, although somewhat larger than, independent estimations of the precision of the MK system. If we tentatively assume that this interpretation is correct, then parameter ξ_1 accounts for 94.5 per cent of the error-free variance of the MK types. Parameter ξ_2 accounts for a further 3.4 per cent, and parameter ξ_5 accounts for a further 1.4 per cent. It is curious that ξ_5 gives a bigger contribution to the variance of the MK types than ξ_3 or ξ_4 , while ξ_5 corresponds to the *smallest* eigenvalue. Table VI shows that, insofar as ξ_5 does help to describe the set of data, its principal contribution is to the variance of x_5 (Ca II K line), and x_3 (Mg *b* lines), although in both cases, the relative contribution is less than 10 per cent. This ties in with what we said before concerning the Ca II K line as a good index of MK types in these stars. Mg and Ca are of course very similar elements, and tend to be similar in their astrophysical behaviour.

5. *Conclusions.*—The last section has shown how an estimate of a classification parameter, based only on the requirement that it should adequately describe the measurements used in determining it, turns out to have a close correspondence with a well-established classification system derived by acknowledged procedures. The formulation, in terms of the original data, of this parameter has been shown to be close to what would have been obtained had we gone ahead in the conventional way and made a least squares fit of MK types to the data. This is important since it was by no means obvious that the correspondence would exist, and is clearly not a logical necessity. What it means is that both the MK system and the parameter derived here are indeed good systems for the classification of this set of stars. Now as far as the MK system is concerned, its adequacy and sufficiency for normal stars was only established after considerable time and effort had been spent. The adequacy of the parameter K_1 follows from the component analysis suggested in this paper. The point of component analysis methods is that they provide considerable insight into the adequacy and sufficiency of the classification system in a very concise way, whereas such insight is difficult and time consuming to obtain in the ordinary way. We have purposely used an example here where most of the “answer” is known already, so that the possible potentialities of the technique could be better appreciated.

To summarize, we have produced, in Table II, quasi-spectral type parameters for a sample of late type giants. A second parameter is indicated as being of statistical significance for describing the set of data, although its relation to the physical properties of the stars, such as absolute magnitude, or chemical composition, is not clear. This paper has been largely concerned with illustrating the technique, and clearly further application of the methods to more virgin fields is called for. The author hopes to consider some of these extensions in later papers.

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